

FLAT TAME WEIGHTS AND VALUATIVE SEPARATION

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ABSTRACT. We prove that a tame maximal weight is flat if and only if its holomorphic relative types define a valuation. Its residual Monge–Ampère mass is then the volume of that valuation. Combined with valuative interpolation and the identification of its obstruction with log discrepancy, this gives a correspondence between flat tame weights modulo bounded functions and centered valuations of finite log discrepancy. Zhou weights are flat as a consequence. Applying the criterion to the quasi-monomial weights constructed by Boucksom–Favre–Jonsson proves the remaining implication (4) $\Rightarrow$ (1) in their Theorem A.

1. INTRODUCTION

Work at the origin $o \in \mathbb{C}^n$, with $n \geq 2$, and put $R = \mathcal{O}_{\mathbb{C}^n, o}$ and $\mathfrak{m} = \mathfrak{m}_o$. For a maximal plurisubharmonic weight Φ with isolated pole at o , write

$$\begin{aligned}\sigma(u, \Phi) &= \sup\{c \geq 0 : u \leq c\Phi + O(1)\}, \\ \nu_\Phi(u) &= (dd^c u \wedge (dd^c \Phi)^{n-1})\{o\}, \quad \tau_\Phi = (dd^c \Phi)^n\{o\}.\end{aligned}$$

Demailly’s comparison theorem gives $\tau_\Phi \sigma(u, \Phi) \leq \nu_\Phi(u)$. Equality for every psh germ is the defining property of a *flat* weight [13, Section 6]. We characterize flatness among tame maximal weights through the function

$$w_\Phi(f) = \sigma(\log |f|, \Phi), \quad f \in R \setminus \{0\}.$$

Theorem 1.1 (Flatness criterion). *Let Φ be a tame maximal weight with isolated pole at o . The following conditions are equivalent:*

- (i) $\nu_\Phi(u) = \tau_\Phi \sigma(u, \Phi)$ for every psh germ u ;
- (ii) w_Φ is a centered real valuation on R .

When these conditions hold, put $w = w_\Phi$, $\lambda = w(\mathfrak{m})$, and $v = w/\lambda$. Then

$$(1.1) \quad \tau_\Phi = \text{vol}(w) > 0, \quad \nu_\Phi(u) = \text{vol}(w) \sigma(u, \Phi) \quad \text{for every psh germ } u.$$

In the normalized valuative space of Boucksom–Favre–Jonsson,

$$(1.2) \quad \widehat{\Phi}(v) = -\lambda^{-1}, \quad \text{MA}(\widehat{\Phi}) = \lambda \text{vol}(w) \delta_v, \quad \sigma(u, \Phi) = \lambda(-\widehat{u}(v)).$$

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For a centered real valuation w , we use the notation

$$\mathfrak{a}_t(w) = \{f \in R : w(f) \geq t\}, \quad \text{vol}(w) = \lim_{m \rightarrow \infty} \frac{n! \ell(R/\mathfrak{a}_m(w))}{m^n}.$$

The main algebraic input is

$$(1.3) \quad \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}} = w(f) \text{vol}(w), \quad 0 \neq f \in \mathfrak{m}.$$

Here $e(I; S)$ denotes Hilbert–Samuel multiplicity. Cutkosky’s theorem [5] supplies the hypersurface limit; the colon identity $(\mathfrak{a}_t(w) : f) = \mathfrak{a}_{t-w(f)}(w)$ determines it by a telescoping sum. Tame approximation transfers the result to analytic masses. The valuative formulas of [4] then extend equality from holomorphic logarithms to all psh germs.

Here and below, a centered real valuation is finite on nonzero germs, zero on units, and strictly positive on $\mathfrak{m} \setminus \{0\}$.

Zhou weights have tame maximal representatives whose holomorphic relative types are valuations [3]. The criterion therefore gives the following consequence; their definition is recalled in [section 5.3](#).

Corollary 1.2 (Zhou weights are flat). *Every local Zhou weight Φ , represented by its tame maximal representative, is flat. If $w = w_\Phi$, then*

$$\tau_\Phi = \text{vol}(w), \quad \nu_\Phi(u) = \text{vol}(w) \sigma(u, \Phi) \quad \text{for every psh germ } u.$$

The notation $\widehat{u}$ denotes the valuative transform of a psh germ in [4]; the sign convention is $\widehat{\log |f|}(v) = -v(f)$.

Theorem 1.3 (Quasi-monomial tests suffice). *For each normalized quasi-monomial valuation v , let Φ_v be the tame maximal weight of [4, Theorem 5.13], characterized by $\sigma(u, \Phi_v) = -\widehat{u}(v)$ for every psh germ u . For psh germs u_1, u_2 at o , the following are equivalent:*

- (i) $\nu_{\Phi_v}(u_1) = \nu_{\Phi_v}(u_2)$ for every $v \in \mathcal{V}_{\text{qm}}$;
- (ii) $\widehat{u}_1 = \widehat{u}_2$ on the normalized valuative space.

In particular, equality of generalized Lelong numbers against all tame weights implies equality of Lelong numbers on all modifications, giving (4) $\Rightarrow$ (1) in Theorem A of [4].

The full range of [theorem 1.1](#) is described in [theorem 5.2](#). The existence theorem of Bao–Guan–Mi–Yuan [2, Theorem 1.11], together with $\mathcal{A} = A$ from [1], identifies flat tame maximal weights modulo bounded functions with all centered real valuations of finite log discrepancy.

2. WEIGHTS AND ANALYTIC INPUTS

2.1. Conventions and logarithmic bounds. We exclude the identically $-\infty$ germ from PSH_o . Use $d^c = (\partial - \bar{\partial})/(2\pi i)$, so that $dd^c \log |f| = [\text{div}(f)]$. For a nonzero ideal $I = (f_1, \dots, f_r)$, set

$$\log |I| = \frac{1}{2} \log(|f_1|^2 + \dots + |f_r|^2).$$

Changing generators changes this germ by a bounded function. All comparisons of singularities in this paper are modulo bounded functions. Weights have isolated poles and continuous exponentials. A weight is maximal if $(dd^c\Phi)^n$ vanishes off o . The masses are the classical generalized Lelong products, not non-pluripolar products. The multiplier ideal is

$$\mathcal{J}(u)_o = \{f \in R : |f|^2 e^{-2u} \text{ is locally integrable at } o\}.$$

We omit the stalk subscript below.

Tameness means that a constant $C > 0$ satisfies

$$(2.1) \quad f \in \mathcal{J}(t\Phi) \implies \log |f| \leq (t - C)\Phi + O(1), \quad t > 0.$$

Lemma 2.1. *A tame weight with isolated pole has constants $0 < c < L$ such that*

$$(2.2) \quad L \log |z| + O(1) \leq \Phi \leq c \log |z| + O(1);$$

In particular, $0 < c^n \leq \tau_\Phi \leq L^n < \infty$.

Proof. Fix $t > C$. The ideal $\mathcal{J}(t\Phi)$ is nonzero by Demailly approximation and is the unit ideal away from o . It is proper at o : otherwise (2.1) applied to 1 would give $0 \leq (t - C)\Phi + O(1)$, contradicting the pole. Coherence therefore makes it $\mathfrak{m}$ -primary, so it contains $\mathfrak{m}^N$ for some N . Apply (2.1) to $z_1^N, \dots, z_n^N$ and take their maximum. This gives the lower bound in (2.2). If the ordinary Lelong number of Φ were zero, Skoda's integrability theorem would make $e^{-2t\Phi}$ locally integrable for every $t > 0$, contradicting properness just proved. The standard logarithmic upper estimate for a psh germ with positive Lelong number gives the upper bound; see [7] for these integrability and Lelong-number facts. Mass comparison gives the final assertion. $\square$

2.2. Comparison, approximation, and multiplicity. We use Demailly's comparison and approximation theorems [6, 7]; the relative-type attainment statement is discussed in [12] and [4, Section 5.4].

Proposition 2.2. *The following statements hold.*

- (i) *For maximal Φ , one has $u \leq \sigma(u, \Phi)\Phi + O(1)$. The relative type is finite for $u \in \text{PSH}_o$ when (2.2) holds.*
- (ii) *For $t > 0$, Demailly approximation gives $u \leq t^{-1} \log |\mathcal{J}(tu)| + O(1)$.*
- (iii) *If weights $\alpha \leq \beta + O(1)$ have isolated poles, then $\nu_\alpha(u) \geq \nu_\beta(u)$. Residual Monge–Ampère masses obey the same monotonicity. Scaling a weight by $b > 0$ multiplies these two quantities by b^{n-1} and b^n , respectively.*
- (iv) *If I is $\mathfrak{m}$ -primary, then*

$$(dd^c \log |I|)^n \{o\} = e(I; R), \quad \nu_{\log |I|}(\log |f|) = e(I; R/(f)) \quad (0 \neq f \in \mathfrak{m}).$$

For irreducible f , Poincaré–Lelong identifies the second mass in (iv) with $(dd^c \log |I|)^{n-1} \wedge [f = 0] \{o\}$. The analytic multiplicity formula on the possibly singular reduced hypersurface is also a case of [8, Theorem 1.1], with the same ideal repeated $n - 1$ times. For general f , both sides are additive over its irreducible factors, with their multiplicities.

2.3. The BFJ valuative formulas. Let $\mathcal{V}$ be the compact space of semi-valuations on the completed local ring, centered at o and normalized by $v(\mathfrak{m}) = 1$. Evaluations of nonzero elements may equal $+\infty$ at a semivaluation. Normalized valuations on R extend uniquely to this space by $\mathfrak{m}$ -adic truncation. Write $\mathcal{V}_{\text{qm}}$ for the quasi-monomial points. For an ideal $I = (f_i)$, $v(I) = \min_i v(f_i)$; in particular $v(IJ) = v(I) + v(J)$. At a divisorial valuation ord_E , $-\widehat{u}(\text{ord}_E)$, interpreted homogeneously, is the generic Lelong number of the pullback along E .

The valuative transform $\widehat{u}$ is a nonpositive formal psh function. It satisfies

$$\widehat{\log |I|}(v) = -v(I), \quad \widehat{bu} = b\widehat{u} \quad (b > 0), \quad \widehat{\max(u_1, u_2)} = \max(\widehat{u}_1, \widehat{u}_2).$$

For an unnormalized real valuation w , put

$$(2.3) \quad w[u] = w(\mathfrak{m})(-\widehat{u}(w/w(\mathfrak{m}))).$$

In particular, $w[\log |I|] = w(I)$.

Theorem 2.3 ([4]). *Let Φ be a tame maximal weight and $g = \widehat{\Phi}$. Then g is bounded, continuous, and strictly negative on $\mathcal{V}$, and*

$$(2.4) \quad \nu_{\Phi}(u) = \int_{\mathcal{V}} -\widehat{u} d\text{MA}(g),$$

$$(2.5) \quad \sigma(u, \Phi) = \inf_{v \in \mathcal{V}} \frac{-\widehat{u}(v)}{-g(v)}.$$

Every formal psh function h admits a sequence $h_j = c_j \log |I_j|$ decreasing to h at every point of $\mathcal{V}$, where $c_j > 0$ and I_j are primary ideals. Finally, its values on all of $\mathcal{V}$ are determined by its values on $\mathcal{V}_{\text{qm}}$:

$$(2.6) \quad h(v) = \inf_{\pi} h(r_{\pi}(v)).$$

Here $r_{\pi}(v)$ is the quasi-monomial retraction associated to a sufficiently high simple modification π .

The sources are Theorem B, Theorem 5.12, Proposition 3.13, and Definition 2.8 of [4], respectively. Boundedness and strict negativity follow from (2.2). Continuity follows from the tame multiplier-ideal approximation [4, Lemma 5.9]: the continuous transforms of primary ideal weights converge uniformly, since their errors are bounded by a constant times $(-g)/m$. Primary ideals in the formal approximation may be regarded as analytic ideals: if $\mathfrak{m}^N \subset I$, truncating generators to polynomials modulo $\mathfrak{m}^N$ does not change the completed ideal or its valuation function.

3. VALUATION IDEALS ON HYPERSURFACES

Let w be any centered real valuation of $R = \mathbb{C}\{z_1, \dots, z_n\}$. Set $\lambda = w(\mathfrak{m}) > 0$ and

$$\mathfrak{a}_t = \{f : w(f) \geq t\} \quad (t > 0), \quad \mathfrak{a}_t = R \quad (t \leq 0), \quad H(t) = \ell_R(R/\mathfrak{a}_t).$$

The family is decreasing and graded: $\mathfrak{a}_s \mathfrak{a}_t \subset \mathfrak{a}_{s+t}$. Moreover $\mathfrak{m}^{\lceil t/\lambda \rceil} \subset \mathfrak{a}_t \subset \mathfrak{m}$ for $t > 0$, so its positive-index ideals are primary.

Theorem 3.1 (Cutkosky). *Let $(S, \mathfrak{n})$ be an analytically unramified equicharacteristic local ring of dimension $d > 0$, with perfect residue field. For a graded family $I_\bullet$ of $\mathfrak{n}$ -primary ideals, both limits below exist and*

$$\lim_{m \rightarrow \infty} \frac{d! \ell_S(S/I_m)}{m^d} = \lim_{m \rightarrow \infty} \frac{e(I_m; S)}{m^d}.$$

This is [5, Theorems 5.8 and 6.5]. Reduced complex analytic local rings are excellent with reduced completion, so the theorem applies to R and to $R/(f)$ for irreducible f . In particular,

$$(3.1) \quad \text{vol}(w) = \lim_{m \rightarrow \infty} \frac{n! H(m)}{m^n} = \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m; R)}{m^n}.$$

The limit is finite; positivity is not assumed here. Monotonicity of H extends its asymptotic formula from integer to real indices by comparison with the floor and ceiling.

Proposition 3.2 (Hypersurface-volume formula). *For $0 \neq f \in \mathfrak{m}$,*

$$\lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}} = w(f) \text{vol}(w).$$

No finite-generation or quasi-monomiality hypothesis is required.

Proof. First suppose f is irreducible. Put $s = w(f) > 0$ and $S = R/(f)$. For every real t and $h \in R$,

$$h \in (\mathfrak{a}_t : f) \iff w(fh) = s + w(h) \geq t \iff h \in \mathfrak{a}_{t-s}.$$

Thus multiplication by f gives the exact sequence

$$(3.2) \quad 0 \longrightarrow R/\mathfrak{a}_{t-s} \xrightarrow{\cdot f} R/\mathfrak{a}_t \longrightarrow R/(\mathfrak{a}_t + (f)) \longrightarrow 0.$$

Consequently

$$(3.3) \quad L_f(t) := \ell_S(S/\mathfrak{a}_t S) = H(t) - H(t-s).$$

The ideals $\mathfrak{a}_m S$ are a graded primary family in the reduced analytic local ring S of dimension $n-1$. By [theorem 3.1](#), there is a finite number B_f such that

$$(3.4) \quad \lim_{m \rightarrow \infty} \frac{(n-1)! L_f(m)}{m^{n-1}} = B_f = \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m; S)}{m^{n-1}}.$$

Since L_f is increasing, its asymptotic formula again holds for real indices. Evaluating (3.3) at $t = s, 2s, \dots, qs$ and using $H(0) = 0$, we obtain

$$\begin{aligned} H(qs) &= \sum_{j=1}^q L_f(js) \\ &= \frac{B_f s^{n-1}}{(n-1)!} \sum_{j=1}^q j^{n-1} + o(q^n) \\ &= \frac{B_f s^{n-1}}{n!} q^n + o(q^n). \end{aligned}$$

Here $\sum_{j \leq q} o(j^{n-1}) = o(q^n)$. On the other hand, (3.1) yields

$$H(qs) = \frac{\text{vol}(w)s^n}{n!} q^n + o(q^n).$$

Comparison gives $B_f = s \text{vol}(w)$, proving the irreducible case.

For general f , unique factorization in R gives $f = a \prod_{i=1}^r f_i^{b_i}$, with a a unit and f_i pairwise nonassociate irreducibles. The associativity formula for multiplicity [10, Chapter 11] gives

$$e(\mathfrak{a}_m; R/(f)) = \sum_{i=1}^r b_i e(\mathfrak{a}_m; R/(f_i)).$$

Indeed, $R_{(f_i)}$ is a discrete valuation ring and $\ell((R/(f))_{(f_i)}) = b_i$. Divide by m^{n-1} and use the irreducible case together with $w(f) = \sum_i b_i w(f_i)$. $\square$

Remark 3.3. The ambient estimate $H(t) = \text{vol}(w)t^n/n! + o(t^n)$ does not by itself determine a fixed-width difference of order t^{n-1} . The existence of the hypersurface limit is essential. Factorization treats the nonreduced case without applying theorem 3.1 to a nonreduced ring.

4. THE FLATNESS CRITERION

Throughout the next two subsections, Φ is a tame maximal weight whose holomorphic relative type $w = w_\Phi$ is a centered real valuation. This is the only valuation hypothesis used in the proof.

4.1. The valuation-ideal approximation.

Lemma 4.1. *There is an integer $C > 0$ independent of m such that, with $\Psi_m = m^{-1} \log |\mathfrak{a}_m(w)|$,*

$$(4.1) \quad (1 + C/m)\Phi \leq \Psi_m \leq \Phi \quad \text{modulo bounded functions.}$$

Proof. Choose an integer C at least the tameness constant. Equation (2.1) implies $\mathcal{J}((m+C)\Phi) \subset \mathfrak{a}_m(w)$. Demailly approximation and inclusion of ideals therefore give

$$(m+C)\Phi \leq \log |\mathcal{J}((m+C)\Phi)| + O(1) \leq \log |\mathfrak{a}_m(w)| + O(1).$$

For the other inequality, each generator f of $\mathfrak{a}_m(w)$ satisfies $\sigma(\log |f|, \Phi) \geq m$. Attainment of relative type implies $\log |f| \leq m\Phi + O(1)$. Taking the maximum over finitely many generators and dividing by m proves (4.1). $\square$

Proposition 4.2. *One has $\tau_\Phi = \text{vol}(w) > 0$ and*

$$(4.2) \quad \nu_\Phi(\log |f|) = \text{vol}(w)w(f) \quad \text{for every } f \in R \setminus \{0\}.$$

Proof. By [proposition 2.2](#) and [lemma 4.1](#),

$$(4.3) \quad \tau_\Phi \leq \frac{e(\mathfrak{a}_m(w); R)}{m^n} \leq (1 + C/m)^n \tau_\Phi.$$

The middle quantity is the residual mass of Ψ_m . Its limit is $\text{vol}(w)$ by (3.1), so the squeeze proves $\tau_\Phi = \text{vol}(w) > 0$, by (2.2).

For irreducible f , the hypersurface multiplicity formula gives

$$\nu_{\Psi_m}(\log |f|) = \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}}.$$

Comparison in the weight variable has degree $n - 1$, hence

$$(4.4) \quad \nu_\Phi(\log |f|) \leq \frac{e(\mathfrak{a}_m(w); R/(f))}{m^{n-1}} \leq (1 + C/m)^{n-1} \nu_\Phi(\log |f|).$$

By [proposition 3.2](#), the middle term tends to $w(f) \text{vol}(w)$. This proves (4.2) for irreducible f . Factorization and additivity of both $dd^c \log |f|$ and $w(f)$ give the general case. A unit contributes zero to both sides. $\square$

4.2. From holomorphic tests to all psh germs.

Lemma 4.3 (Holomorphic testing). *Let Φ be tame and maximal, and suppose*

$$\nu_\Phi(\log |f|) = \tau_\Phi \sigma(\log |f|, \Phi) \quad \text{for every nonzero } f \in R.$$

Then there is a unique $v_0 \in \mathcal{V}$ such that

$$(4.5) \quad \text{MA}(g) = \frac{\tau_\Phi}{-g(v_0)} \delta_{v_0}, \quad \frac{\nu_\Phi(u)}{\tau_\Phi} = \sigma(u, \Phi) = \frac{-\widehat{u}(v_0)}{-g(v_0)}, \quad g = \widehat{\Phi},$$

for every psh germ u .

Proof. Applying (2.4) to $u = \Phi$ gives $\int_{\mathcal{V}} (-g) d\text{MA}(g) = \tau_\Phi$. Thus

$$dP = \frac{-g}{\tau_\Phi} d\text{MA}(g)$$

is a Radon probability measure. For a nonzero holomorphic germ f , set $R_f(v) = v(f)/(-g(v))$ and $a_f = \sigma(\log |f|, \Phi)$. The hypotheses and [theorem 2.3](#) give

$$\inf_{\mathcal{V}} R_f = a_f = \int_{\mathcal{V}} R_f dP.$$

The nonnegative function $R_f - a_f$ therefore has integral zero. It is continuous with values in $[0, +\infty]$, because evaluation of f is continuous in the valuation topology and $-g$ is continuous and bounded away from zero. If $R_f(v) > a_f$

at a point of $\text{supp } P$, some neighborhood of that point has positive P -mass and $R_f - a_f \geq \varepsilon > 0$ there, a contradiction. Consequently

$$(4.6) \quad \frac{v(f)}{-g(v)} = a_f \quad \text{for every } v \in \text{supp } P \text{ and every } f \neq 0.$$

The support is nonempty. If v_1, v_2 belong to it, then $v_1/(-g(v_1))$ and $v_2/(-g(v_2))$ agree on every holomorphic germ. Taking their minima on the coordinate generators of $\mathfrak{m}$ and using $v_i(\mathfrak{m}) = 1$ shows $-g(v_1) = -g(v_2)$, and therefore equality on R . Polynomial truncations determine each normalized semivaluation on the completion: for a formal series F and its degree- N truncation F_N , $v(F - F_N) \geq N + 1$. Hence $v_1 = v_2$ on the completion as well. By Radon regularity, $P = \delta_{v_0}$ for one point v_0 .

It remains to identify the relative type for an arbitrary u . By (4.6) and the definition of the infimum,

$$\frac{v_0(f)}{-g(v_0)} \leq \frac{v(f)}{-g(v)} \quad \text{for every } v \in \mathcal{V} \text{ and every } f \neq 0.$$

Taking the minimum over generators proves this inequality for any primary ideal, and positive scaling preserves it. Choose $h_j = c_j \log |I_j| \downarrow \hat{u}$ as in [theorem 2.3](#). For each fixed v , pass to the limit in

$$\frac{-h_j(v_0)}{-g(v_0)} \leq \frac{-h_j(v)}{-g(v)}.$$

It follows that v_0 attains the infimum in (2.5) for u . Integration against $P = \delta_{v_0}$ gives the generalized Lelong number in (4.5). The measure formula follows from the definition of P . $\square$

Proof of theorem 1.1. First suppose Φ is flat. Linearity in the first argument gives

$$w_\Phi(fg) = \frac{\nu_\Phi(\log |fg|)}{\tau_\Phi} = \frac{\nu_\Phi(\log |f|) + \nu_\Phi(\log |g|)}{\tau_\Phi} = w_\Phi(f) + w_\Phi(g).$$

Relative type always satisfies $\sigma(\max\{u_1, u_2\}, \Phi) = \min_i \sigma(u_i, \Phi)$. Combining this rule with $\log |f + g| \leq \max\{\log |f|, \log |g|\} + \log 2$ gives

$$w_\Phi(f + g) \geq \min\{w_\Phi(f), w_\Phi(g)\}.$$

Set $w_\Phi(0) = +\infty$ to include $f + g = 0$. For a nonzero germ of vanishing order k , the lower logarithmic bound in (2.2) implies $w_\Phi(f) \geq k/L$. Conversely, $\log |f| \leq a\Phi + O(1)$ implies $k \geq ac$ by the upper logarithmic bound and comparison of ordinary Lelong numbers. Thus $k/L \leq w_\Phi(f) \leq k/c$. Units have relative type zero. These facts prove that w_Φ is a centered real valuation.

Conversely, suppose w_Φ is such a valuation. By [proposition 4.2](#), the hypothesis of [lemma 4.3](#) holds. Hence $\nu_\Phi(u) = \tau_\Phi \sigma(u, \Phi)$ for all u . On holomorphic germs, that lemma also gives

$$w(f) = \frac{v_0(f)}{-g(v_0)}.$$

Taking minima on generators of $\mathfrak{m}$ yields $\lambda = 1/(-g(v_0))$, and therefore $v_0 = w/\lambda$. Since $\tau_\Phi = \text{vol}(w)$, all formulas in (1.1) and (1.2) follow. $\square$

Since $\mathfrak{a}_t(bv) = \mathfrak{a}_{t/b}(v)$, one has $\text{vol}(bv) = b^{-n} \text{vol}(v)$. Thus the normalized formulas are

$$\tau_\Phi = \lambda^{-n} \text{vol}(v), \quad \text{MA}(\widehat{\Phi}) = \lambda^{1-n} \text{vol}(v) \delta_v.$$

The squeezes (4.3) and (4.4) give relative errors $O(m^{-1})$ in the corresponding multiplicity limits.

5. FINITE-DISCREPANCY VALUATIONS AND FLAT TAME WEIGHTS

5.1. The interpolation obstruction. For a nonzero germ f , write

$$c^f(\psi) = \sup\{t \geq 0 : |f|^2 e^{-2t\psi} \text{ is locally integrable}\}.$$

For a graded sequence of nonzero ideals $I_\bullet$, put

$$\text{lct}^{(f)}(I_\bullet) = \sup_{m \geq 1} m c^f(\log |I_m|).$$

These are the analytic jumping numbers, equal to their algebraic counterparts on R . Let $A(w)$ denote log discrepancy [11, Section 5]: on divisorial valuations, $A(\text{ord}_E) = 1 + \text{ord}_E(K_{Y/X})$; it is homogeneous and linear on quasi-monomial cones, and for general valuations it is the supremum over quasi-monomial retractions. Its value is allowed to be infinite.

The obstruction in valuative interpolation is

$$(5.1) \quad \mathcal{A}(w) = \sup_{0 \neq f \in R} (\text{lct}^{(f)}(\mathfrak{a}_\bullet(w)) - w(f)).$$

Equivalently, $\mathcal{A}$ can be defined using arbitrary graded sequences and auxiliary ideals; see [1]. We use the following results.

Theorem 5.1 (Valuative interpolation and discrepancy). *Let w be a centered real valuation of R .*

- (i) *The condition $\mathcal{A}(w) < \infty$ is equivalent to the existence of a tame maximal weight Φ_w satisfying*

$$w(f) = \sigma(\log |f|, \Phi_w) \quad (f \neq 0).$$

When this holds, one also has

$$c^f(\Phi_w) = \text{lct}^{(f)}(\mathfrak{a}_\bullet(w)) \quad (f \neq 0).$$

- (ii) *One has $\mathcal{A}(w) = A(w)$, including the value $+\infty$.*

Part (i) is [2, Theorem 1.11]; part (ii) is [1, Theorem 1.1]. An estimate for the continuity of the representing envelope is proved in [appendix A](#).

The inequality $\mathcal{A}(w) \leq A(w)$ proves the existence of a representing weight when $A(w) < \infty$. Indeed, $w(\mathfrak{a}_\bullet(w)) = 1$: the lower bound is immediate, and if $w(f) = s > 0$, the germ $f^{\lceil m/s \rceil}$ has value less than $m + s$ and belongs to $\mathfrak{a}_m(w)$. The valuative threshold inequality [11, Corollary 6.9] consequently gives

$$\text{lct}^{(f)}(\mathfrak{a}_\bullet(w)) \leq A(w) + w(f).$$

5.2. The correspondence.

Theorem 5.2. *The assignment*

$$[\Phi] \mapsto w_\Phi, \quad w_\Phi(f) = \sigma(\log |f|, \Phi),$$

is a bijection from tame flat maximal weights with isolated pole at o , modulo bounded functions, to centered real valuations w with $A(w) < \infty$. The inverse is the weight Φ_w of [theorem 5.1](#). For every $u \in \text{PSH}_o$,

$$\tau_{\Phi_w} = \text{vol}(w) > 0, \quad \sigma(u, \Phi_w) = w[u], \quad \nu_{\Phi_w}(u) = \text{vol}(w)w[u].$$

Replacing Φ by $b\Phi$ corresponds to replacing w_Φ by w_Φ/b , for $b > 0$.

Proof. If Φ is tame, flat, and maximal, [theorem 1.1](#) makes w_Φ a centered real valuation. Its tame representation gives $\mathcal{A}(w_\Phi) < \infty$ by [theorem 5.1\(i\)](#), and part (ii) yields $A(w_\Phi) < \infty$.

Conversely, if $A(w) < \infty$, then $\mathcal{A}(w) < \infty$. Part (i) provides a tame maximal Φ_w with holomorphic relative type w . The criterion makes it flat and gives all displayed formulas, using the definition [\(2.3\)](#).

It remains to prove uniqueness modulo bounded functions. Suppose Φ_1, Φ_2 are tame maximal weights defining the same w . Their valuation-ideal approximants Ψ_m are identical. By [lemma 4.1](#), for suitable constants C_1, C_2 ,

$$(1 + C_i/m)\Phi_i \leq \Psi_m \leq \Phi_i \quad (i = 1, 2)$$

modulo bounded functions. Combining the lower inequality for $i = 1$ with the upper inequality for $i = 2$ gives

$$\Phi_1 \leq \frac{1}{1 + C_1/m} \Phi_2 + O(1).$$

Hence $\sigma(\Phi_1, \Phi_2) \geq (1 + C_1/m)^{-1}$ for every m , and therefore $\sigma(\Phi_1, \Phi_2) \geq 1$. Attainment for the maximal weight Φ_2 yields $\Phi_1 \leq \Phi_2 + O(1)$. Interchanging the indices gives the reverse inequality, so $\Phi_1 = \Phi_2 + O(1)$. Finally, the scaling statement follows directly from the definition of relative type. $\square$

5.3. Zhou weights. For completeness, recall the defining extremality. Let F_0 be a nonzero holomorphic tuple and ψ_0 a psh germ such that $|F_0|^2 e^{-2\psi_0}$ is locally integrable. A local Zhou weight Φ for this datum satisfies:

- (i) $|F_0|^2 e^{-2\psi_0} |z|^{2N} e^{-2\Phi}$ is integrable for some N ;
- (ii) $|F_0|^2 e^{-2\psi_0} e^{-2\Phi}$ is not integrable;
- (iii) if $\Psi \geq \Phi + O(1)$ is psh and the expression in (ii) with Ψ in place of Φ is not integrable, then $\Psi = \Phi + O(1)$.

Bao–Guan–Mi–Yuan [\[3, Section 1.1, Proposition 1.7, and Section 1.2\]](#) prove that these weights admit tame maximal representatives and that their holomorphic relative types are centered real valuations.

Proof of [corollary 1.2](#). Apply [theorem 1.1](#) to the two properties from [\[3\]](#) just stated. $\square$

6. QUASI-MONOMIAL TESTS AND THE BFJ CONVERSE

6.1. Quasi-monomial weights. The construction in [4, Theorem 5.13] gives, for each normalized quasi-monomial v , a tame maximal weight Φ_v satisfying

$$(6.1) \quad \sigma(u, \Phi_v) = -\widehat{u}(v) \quad (u \in \text{PSH}_o).$$

On taking $u = \log |f|$, its holomorphic relative type is exactly v . Consequently [theorem 1.1](#), with $\lambda = 1$, gives

$$(6.2) \quad \tau_{\Phi_v} = \text{vol}(v) > 0, \quad \nu_{\Phi_v}(u) = \text{vol}(v)(-\widehat{u}(v)).$$

Thus these generalized Lelong tests are positive multiples of point evaluations.

Proof of [theorem 1.3](#). Suppose all tests in condition (i) agree. By (6.2), division by the same positive number $\text{vol}(v)$ gives $\widehat{u}_1(v) = \widehat{u}_2(v)$ for every $v \in \mathcal{V}_{\text{qm}}$. For any $\mu \in \mathcal{V}$, the retraction formula (2.6) gives

$$\widehat{u}_1(\mu) = \inf_{\pi} \widehat{u}_1(r_{\pi}(\mu)) = \inf_{\pi} \widehat{u}_2(r_{\pi}(\mu)) = \widehat{u}_2(\mu).$$

This proves (ii). Conversely, equality of the valuative transforms implies equality in (6.2), and in fact equality of the integrals (2.4) against every tame weight.

If condition (4) of Theorem A in [4] holds, its tests include all Φ_v . The implication just proved gives equality of valuative transforms. By [4, Sections 5.1 and 5.6], this is equivalent to equality of all infinitely near Lelong numbers, which is their condition (1). $\square$

Even the divisorial subfamily of (6.1) suffices. Indeed, on each quasi-monomial simplex, rational points are normalized divisorial valuations and are dense, while the restrictions of $\widehat{u}_i$ are finite and continuous [4, Proposition 5.6]. Equality at these rational points therefore gives equality throughout every simplex. The retraction formula (2.6) gives equality on $\mathcal{V}$.

Corollary 6.1 (Order detection). *For $u_1, u_2 \in \text{PSH}_o$, one has $\widehat{u}_1 \leq \widehat{u}_2$ on $\mathcal{V}$ if and only if*

$$\nu_{\Phi_v}(u_1) \geq \nu_{\Phi_v}(u_2) \quad \text{for every normalized quasi-monomial } v.$$

Proof. Use (6.2) and then (2.6); conversely, integrate $-\widehat{u}_1 \geq -\widehat{u}_2$ against $\text{MA}(\widehat{\Phi}_v)$. $\square$

Corollary 6.2 (Multiplier ideals). *Under either condition of [theorem 1.3](#), $\mathcal{J}(tu_1) = \mathcal{J}(tu_2)$ for every $t > 0$.*

Proof. The implication (1) $\Rightarrow$ (2) in [4, Theorem A] gives equality of $\mathcal{J}_+(tu_i)$. Strong openness [9] identifies these with $\mathcal{J}(tu_i)$. $\square$

6.2. Separation by Zhou weights.

Corollary 6.3. *For $u_1, u_2 \in \text{PSH}_o$, equality $\nu_\Phi(u_1) = \nu_\Phi(u_2)$ for every local Zhou weight Φ is equivalent to $\hat{u}_1 = \hat{u}_2$.*

Proof. By [corollary 1.2](#), equality of these masses gives equality of the relative types against all Zhou weights, after division by $\tau_\Phi > 0$. The relative-type characterization of multiplier ideals in [\[3\]](#) then gives $\mathcal{J}(tu_1) = \mathcal{J}(tu_2)$ for every $t > 0$. Strong openness identifies these ideals with $\mathcal{J}_+(tu_i)$. The implication (2) $\Rightarrow$ (1) in [\[4, Theorem A\]](#) gives equality of valuative transforms. The converse follows from [\(2.4\)](#). $\square$

In dimension one, the conclusions follow directly from $\Phi = a \log |z| + O(1)$: its residual mass is a , and both its normalized generalized Lelong number and its relative type equal $\nu(u, o)/a$.

APPENDIX A. A CONTINUITY ESTIMATE FOR THE INTERPOLATION ENVELOPE

We prove a local uniform estimate for the Bergman approximations of the envelope in [\[2, Lemma 2.32\]](#).

Let Φ denote the negative envelope called φ_2 in [\[2, Section 2.5\]](#), on a bounded hyperconvex domain D containing o . The function Φ is locally bounded on $D \setminus \{o\}$, satisfies logarithmic bounds, and tends to zero at ∂D . The boundary limit follows from the exhaustion-barrier argument in its [Lemma 2.32](#). [Lemmas 2.29–2.31](#) there also give a constant $C > 0$ with

$$(A.1) \quad c^f(\Phi) - \sigma(\log |f|, \Phi) \leq C \quad (f \neq 0),$$

and the following envelope property. For $t > 0$, let $D_t = \{\Phi < -t\}$. The corresponding envelope on D_t is $\Phi + t$, and every negative psh function h on D_t satisfies

$$(A.2) \quad h \leq \sigma(h, \Phi)(\Phi + t).$$

Here the relative type is unchanged upon adding the constant t .

For the Hilbert space

$$\mathcal{H}_m = \left\{ f \in \mathcal{O}(D) : \int_D |f|^2 e^{-2m\Phi} dV < \infty \right\},$$

let ψ_m be its Bergman approximation:

$$\psi_m(z) = \frac{1}{m} \log \sup_{\|f\|_{\mathcal{H}_m} \leq 1} |f(z)|.$$

Demailly's estimate gives $\psi_m \geq \Phi - c_1/m$, with c_1 independent of m . We claim that for every fixed $t > 0$ there is $B_t > 1$ such that

$$(A.3) \quad \Phi - \frac{c_1}{m} \leq \psi_m \leq \left(1 - \frac{C}{m}\right)(\Phi + t) + \frac{\log B_t}{m} \quad \text{on } D_t, \quad m > C.$$

Indeed, $\overline{D_t}$ is compact in D by the boundary limit. For $\|f\|_{\mathcal{H}_m} \leq 1$, negativity of Φ implies $\int_D |f|^2 dV \leq 1$. The ordinary interior mean-value

estimate, on balls of a fixed radius smaller than $\text{dist}(\overline{D}_t, \partial D)$, consequently gives $|f| < B_t$ on D_t , independently of m and f . For $f \neq 0$, the function $h = \log(|f|/B_t)$ is negative and psh there. Its germ satisfies $c^f(\Phi) \geq m$, so (A.1) gives $\sigma(h, \Phi) \geq m - C$. Since $\Phi + t < 0$, (A.2) yields

$$\log |f| - \log B_t \leq \sigma(h, \Phi)(\Phi + t) \leq (m - C)(\Phi + t).$$

Taking the supremum over the unit ball proves the upper estimate in (A.3).

To conclude, fix $K \Subset D \setminus \{o\}$. Since $\max_K \Phi < 0$, all sufficiently small $t > 0$ satisfy $K \subset D_t$. The function Φ is bounded on K , and (A.3) implies

$$\limsup_{m \rightarrow \infty} \|\psi_m - \Phi\|_{L^\infty(K)} \leq t.$$

Let $t \downarrow 0$. Thus $\psi_m \rightarrow \Phi$ locally uniformly off o . The Bergman kernel is smooth and positive there: smoothness follows from local convergence of the holomorphic kernel series, and positivity from Demailly's lower estimate. Hence Φ is continuous off o . Its logarithmic upper bound implies $e^\Phi \rightarrow 0$ at o , so e^Φ is continuous on D ; at ∂D it tends to one. Together with (A.1), this is the required tame maximal representative.

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DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

The author proposed the research questions. The proofs and exposition were developed with GPT-6 Astra and GPT-5.6-sol (high), and remain subject to the author's mathematical verification.

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